

Research Methodology in CS (CSEG3060_3)

Lecture 7: Patenting under PCT (Patent Cooperation Treaty): Structure and objectives of PCT, phases of PCT application (international phase and national phase), advantages of PCT route, timelines, fees, and procedural requirements for obtaining patent protection in multiple countries.

Unit 5: Intellectual Property and Patents • Mapped Course Outcome: CO3

Lecture 7: Patenting under the Patent Cooperation Treaty (PCT)

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1. Introduction

The globalized research and innovation ecosystem demands mechanisms that allow inventors, researchers, and technology companies to seek patent protection across multiple jurisdictions efficiently. Filing separate patent applications in every country of interest is a cumbersome, expensive, and time-consuming process. The **Patent Cooperation Treaty (PCT)**, administered by the World Intellectual Property Organization (WIPO), addresses this challenge by offering a unified procedural framework for filing, searching, and examining patent applications internationally.

This lecture explores the **structure, objectives, procedural architecture, advantages, timelines, fees, and requirements** of the PCT system. A thorough understanding of the PCT route is essential for computer science researchers engaged in software-related inventions, algorithms, AI models, hardware innovations, and interdisciplinary technologies where multi-jurisdictional protection is often desirable.

2. Historical Background and Genesis of PCT

The PCT was signed in **Washington, D.C., in 1970** and came into force on **January 24, 1978**. The treaty was the culmination of years of deliberations among WIPO member states to harmonize patent filing procedures and reduce duplication of effort among national and regional patent offices.

Before the PCT, inventors relied primarily on the **Paris Convention for the Protection of Industrial Property (1883)**, which provided a right of priority within 12 months but required separate filings in each country. The PCT builds upon this convention by extending the priority period and providing an internationally coordinated search and preliminary examination procedure.

As of recent years, the PCT has more than **150 contracting states**, making it one of the most successful international treaties in the intellectual property domain.

3. Structure of the PCT System

The PCT operates through a coordinated network of offices and authorities:

3.1 Key Bodies and Offices

1. **International Bureau (IB)**: Housed within WIPO in Geneva, it administers the treaty, publishes international applications, and coordinates operations.
2. **Receiving Office (RO)**: The national or regional patent office (or WIPO itself) where the PCT application is initially filed.
3. **International Searching Authority (ISA)**: Conducts prior art search and issues the International Search Report (ISR) and Written Opinion. Examples include the USPTO, EPO, JPO, KIPO, and the Indian Patent Office.
4. **International Preliminary Examining Authority (IPEA)**: Performs an optional, non-binding preliminary examination on patentability, resulting in an International Preliminary Report on Patentability (IPRP).
5. **Designated Offices (DO) / Elected Offices (EO)**: The national or regional patent offices of contracting states where the applicant ultimately seeks protection.

3.2 Structural Highlights

- The PCT is **not a granting treaty**; it does not itself issue patents.
- It functions as a **filing and procedural harmonization system**.

- The treaty consists of **Articles (69 articles)** and **Rules (96 rules)**, supplemented by the Administrative Instructions.
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4. Objectives of the PCT

The principal objectives of the PCT, as articulated in Article 1, are:

1. **Unified Filing Procedure:** To allow a single international application to have the effect of a regular national filing in each contracting state.
 2. **Avoidance of Duplication of Effort:** To streamline the search and examination process so that inventors do not file and prosecute identical applications in multiple countries simultaneously.
 3. **Extended Priority Period:** To defer substantial national filing costs by providing an additional **18 months** beyond the Paris Convention's 12-month priority period (totaling **30 months** from the priority date) before national phase entry is required.
 4. **Information Dissemination:** To publish international applications and search reports, providing technical information to the public and industry.
 5. **Facilitating Technology Transfer:** To assist developing countries by making technical information accessible and supporting their patent examination capabilities.
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5. The Two Phases of a PCT Application

The PCT procedure is structured into two distinct phases:

5.1 International Phase

The international phase encompasses all activities from filing of the PCT application up to (but excluding) the commencement of processing before individual national or regional patent offices.

5.1.1 Filing of the International Application

The applicant files a single application at a competent Receiving Office. The application comprises:

- A **request** (Form PCT/RO/101 or equivalent ePCT filing)
- A **description** of the invention
- One or more **claims**
- **Drawings** (if applicable)

- An **abstract**

The filing date becomes the **international filing date**, provided the application meets minimum requirements under PCT Article 11.

5.1.2 International Search

After approximately **16 months** from the priority date, the ISA conducts a comprehensive prior art search and issues:

- **International Search Report (ISR)**: Lists relevant prior art documents classified according to international patent classification codes.
- **Written Opinion of the ISA**: Preliminary non-binding opinion on novelty, inventive step (non-obviousness), and industrial applicability.

The applicant has **two months** (or **three months** from the search report's mailing date) to amend claims before publication. This opportunity allows the inventor to refine the scope of protection in light of the prior art discovered.

5.1.3 International Publication

The international application is published by the International Bureau approximately **18 months** from the priority date. Publication includes the application as filed, any amended claims, the ISR, and (later) the IPRP. Publication confers **provisional protection** in designated states.

5.1.4 Supplementary International Search (Optional)

An applicant may request a supplementary international search by a different ISA to cover prior art not retrieved in the main search (e.g., documents in non-English languages or specialized databases).

5.1.5 International Preliminary Examination (Optional but Strategically Important)

Under **Chapter II** of the PCT, the applicant may request an international preliminary examination. The IPEA reviews the application and issues an **International Preliminary Report on Patentability (IPRP Chapter II)**. This report, although non-binding on national offices, carries persuasive weight and helps applicants:

- Assess the likely strength of their patent before incurring national phase costs.
- Amend the application to address substantive objections.
- Make informed decisions about entering national phases.

The request for examination must be filed within **22 months** from the priority date (in some jurisdictions up to 25 months for certain Offices).

5.2 National Phase

The national phase begins when the applicant files the application with individual designated (or elected) Offices. Critical aspects include:

- **Deadline:** Generally **30 months** from the priority date (some Offices allow **31 months**, e.g., Australia, Europe).
- **Translations:** The application must be translated into the official language(s) of each designated state.
- **National Fees:** Payment of filing fees, examination fees, and annuities applicable in each jurisdiction.
- **Representation:** Engagement of local patent attorneys/agents.
- **National Examination:** Each national office examines the application according to its substantive patent laws.

In the national phase, the previously conducted international search and examination may be utilized; some offices (e.g., EPO, JPO) provide accelerated examination using PCT results.

6. Advantages of the PCT Route

6.1 Procedural Advantages

1. **Single Initial Filing:** One application substitutes for potentially dozens of individual national filings.
2. **Extended Decision Window:** The applicant gains **18 additional months** (30 months total from priority) before committing to national phase expenses.
3. **Preliminary Patentability Assessment:** The ISR, Written Opinion, and optional IPRP enable applicants to evaluate the commercial viability of pursuing protection in specific countries.
4. **Deferred Translation Costs:** Translations into national languages are required only at national phase entry, allowing resources to be allocated strategically.
5. **Publication Benefits:** International publication provides early public disclosure, establishes priority, and may attract licensing or investment interest.

6.2 Strategic Advantages

- 1. **Risk Mitigation:** The option to abandon the application before national phase entry saves substantial costs where prospects are weak.
- 2. **Market Flexibility:** The extended timeline allows the applicant to assess market conditions, manufacturing feasibility, and competitor activity before committing.
- 3. **Search Quality:** International searches leverage comprehensive databases across multiple languages.
- 4. **Examination Coordination:** Many national offices cooperate with PCT results, expediting examination.

6.3 Comparison with Other Routes

Feature	Direct National Filing	Paris Convention Route	PCT Route
Filing Cost	High in each country	Multiple filings needed	Single initial filing
Search Quality	National only	National only	International (multiple ISAs)
Decision Window	None	12 months	30 months
Translation	Immediate	Immediate	Deferred to national phase
Procedural Complexity	High (multiple filings)	Moderate	Streamlined
Global Reach	Limited by national effort	Better but still costly	Highest (150+ countries)

7. Timelines and Critical Deadlines

A clear understanding of timelines is essential for effective PCT prosecution.

7.1 Key Milestones (Measured from Priority Date)

Milestone	Timeline
Priority date established	0 months
International application filing (PCT)	Up to 12 months
International search report and written opinion	~16 months
International publication	~18 months
Deadline for Chapter II demand	22 months (varies by Office)
National phase entry deadline	30 months (varies: 31 in some states)
Restoration opportunity (in some Offices)	After 30 months (subject to conditions)

7.2 Restoration and Extensions

- **PCT Rule 26bis.1 / 26bis.3:** Restoration of right of priority and restoration of international application, applicable in designated Offices.
- Many Offices allow **late national phase entry** under specific conditions with additional fees.

8. Fees Structure

PCT fees comprise three categories:

8.1 International Phase Fees

1. **Transmittal Fee:** Charged by the Receiving Office (varies by Office).
2. **International Filing Fee:** Approximately **CHF 1,330** (for the first 30 sheets; reductions apply for online filings).
3. **Search Fee:** Charged by the chosen ISA (ranges from CHF 200 to CHF 2,000 depending on ISA).
4. **Preliminary Examination Fee:** Charged by the IPEA under Chapter II.
5. **Supplementary International Search Fee:** Optional additional search.

8.2 National Phase Fees

- Vary significantly by jurisdiction.
- Include filing fees, examination fees, grant fees, and annuities.

- For a typical 5-country filing, national phase costs may range from **USD 15,000 to USD 80,000** including attorney fees and translations.

8.3 Fee Reductions

- **75% reduction** on international filing, search, and preliminary examination fees for natural persons and small entities from eligible developing countries.
 - **90% reduction** for academic institutions in certain states.
 - Online filing discounts (e.g., CHF 200 reduction on filing fee via ePCT).
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9. Procedural Requirements for Obtaining Multi-Country Protection

9.1 Requirements Before Filing

1. **Establish Priority:** Identify the earliest disclosure (own or otherwise) to determine the priority date.
2. **Draft a Strong Application:** Comprehensive description, well-formulated claims, and suitable drawings.
3. **Choose ISA Strategically:** Select ISA based on technical expertise, language, and cost.

9.2 Requirements During International Phase

1. **Respond to Written Opinion:** Amend claims, arguments, or specification to address objections.
2. **Consider Chapter II:** Request preliminary examination to strengthen prosecution.
3. **Monitor Publication:** Be prepared for public disclosure at 18 months.

9.3 Requirements for National Phase Entry

1. **Translate the Application:** Into official languages of designated states.
 2. **Pay National Fees:** Filing, examination, and annuity fees.
 3. **Appoint Local Representatives:** Engage patent attorneys in each jurisdiction.
 4. **Submit Required Forms:** Country-specific declarations and formal documents.
 5. **Respond to National Office Actions:** Address any further objections under national law.
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10. PCT and Computer Science Inventions

The PCT route is particularly relevant for computer science innovations:

- **Software-related inventions** face varying patentability standards worldwide; PCT allows assessment across jurisdictions.
 - **AI and machine learning inventions** benefit from international prior art searches.
 - **Hardware innovations** (chips, IoT devices) require multi-country protection due to global manufacturing chains.
 - **Business method patents** under PCT face scrutiny per national standards (e.g., USPTO Alice test, EPC exclusions).
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11. Practical Example / Case Study

Case: A CS Research Group Inventing a Novel Federated Learning Algorithm

A university research team in India develops a novel federated learning algorithm with privacy-preserving enhancements. The team wishes to protect the invention in the US, Europe, China, Japan, and South Korea.

Without PCT:

- File provisional in India (priority date).
- Within 12 months, file in 5 jurisdictions with separate translations, attorneys, and fees.
Estimated cost: USD 40,000-60,000.

With PCT Route:

- File Indian provisional (priority date: Day 0).
- File PCT application at Indian Patent Office (acting as Receiving Office) by Month 12.
- ISA = Indian Patent Office or EPO conducts search by Month 16.
- Publication by WIPO at Month 18.
- Optional Chapter II examination between Months 19-22.
- Decision to enter national phase in selected countries by Month 30.
- Cost: International phase ~USD 4,000-6,000; National phase USD 25,000-50,000 (deferred).

The PCT route provided the team an additional 18 months to publish results, secure industry partners, and validate commercial interest before incurring the substantial costs of multi-country patent prosecution.

12. Summary

The Patent Cooperation Treaty represents the most significant procedural harmonization in international patent law. By enabling a single international application with extended timelines, comprehensive prior art searches, and optional preliminary examination, the PCT empowers inventors—particularly those in fast-moving technological fields such as computer science—to make informed, strategic decisions about multi-jurisdictional patent protection.

Key takeaways from this lecture:

- The PCT is **not a granting treaty** but a procedural coordination system.
 - It operates through **two phases**: international and national.
 - It offers a **30-month window** from priority date for strategic decision-making.
 - **Advantages** include cost deferral, search quality, and procedural simplicity.
 - **Fees and timelines** must be carefully managed to avoid loss of rights.
 - For CS innovations, the PCT route is often indispensable for global protection.
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13. Exam Preparation Tips

13.1 Key Topics to Focus On

1. Differentiate between the **international phase** and **national phase**.
2. List the key **objectives of the PCT**.
3. Explain the **role of ISA, IPEA, Receiving Office, and International Bureau**.
4. State the **priority deadline** for PCT filing and **national phase entry deadline**.
5. Discuss the **advantages of the PCT route** over direct national and Paris Convention routes.
6. Describe **fee components** of the international phase.
7. Explain the **non-binding nature** of the ISR and Written Opinion.

13.2 Practice Questions

1. **Short Answer:** What is the role of the International Searching Authority, and how does its report influence subsequent prosecution?
2. **Descriptive:** Discuss the structural objectives of the PCT and explain how they benefit inventors seeking multi-country protection.
3. **Comparative:** Compare and contrast the PCT route with the Paris Convention route for international patent filing. Which is more advantageous for a CS startup, and why?

4. **Numerical/Applied:** An inventor files a provisional application on January 1, 2024. Calculate:
- (a) The last date to file a PCT application claiming priority.
 - (b) The approximate date of international publication.
 - (c) The deadline for national phase entry.
5. **Case Analysis:** A multinational technology company has developed a novel AI-based cybersecurity solution. Outline a PCT-based strategy for securing patent protection in the US, EU, China, Japan, and South Korea. Include timelines, fees, and procedural considerations.
6. **Conceptual:** Why is the PCT not considered a granting treaty? What implications does this have for applicants?
7. **Critical Thinking:** Discuss the strategic value of opting for Chapter II preliminary examination. Under what circumstances should an applicant skip it?
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14. References and Further Reading

1. **WIPO PCT Website:** <https://www.wipo.int/pct/> — Official source for treaty text, forms, and fees.
 2. **PCT Applicant's Guide:** WIPO Publication — Comprehensive procedural guidance.
 3. **Paris Convention for the Protection of Industrial Property (1883)** — Foundational treaty underlying PCT priority.
 4. **World Intellectual Property Indicators (WIPO Annual Report)** — Statistics on PCT filings.
 5. **Indian Patent Act, 1970 (as amended)** — National legislation relevant to PCT national phase.
 6. **European Patent Convention (EPC)** — Regional framework for European patent grants.
 7. **Manual of Patent Examining Procedure (MPEP)** — USPTO reference for PCT procedures.
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End of Lecture 7 Notes